

Global Market Correlation Analysis: Investigating the Relationship Between NIFTY 50 and Major Financial Assets Using Python

Introduction

Financial markets are interconnected through global economic events, capital flows, monetary policies, and investor sentiment. Understanding these relationships is crucial for investors and financial analysts seeking to make informed decisions. This project explores the correlation between the NIFTY 50 index and various global financial assets, including international stock indices, commodities, cryptocurrencies, currency indices, and futures markets. Using Python and Yahoo Finance data, historical market data was collected, analyzed, and visualized to identify trends, performance patterns, and inter-market relationships over a five-year period.

Project Objectives

To collect and process historical financial market data from Yahoo Finance using Python. To compare the performance of NIFTY 50 with major global stock indices such as Nikkei 225, KOSPI, Hang Seng, DAX, FTSE, Nasdaq, and Dow Futures. To analyze the relationship between NIFTY and alternative asset classes including Gold, Silver, Crude Oil, Bitcoin, and the US Dollar Index. To calculate correlation coefficients and identify the degree of association between Indian and global financial markets. To develop data visualizations and analytical dashboards that support investment research and market analysis.

Conclusion

The project demonstrated that the Indian stock market exhibits varying levels of correlation with global financial assets. Equity markets with strong international exposure showed relatively higher relationships with NIFTY, while commodities and cryptocurrencies displayed distinct market dynamics. The analysis highlighted the influence of macroeconomic indicators such as crude oil prices and the US Dollar Index on Indian equities. By leveraging Python-based data analytics and visualization techniques, the project provided valuable insights into global market interdependence and strengthened financial research capabilities.

```
# Install libraries (Colab)
!pip install yfinance --quiet

import yfinance as yf
import pandas as pd
import matplotlib.pyplot as plt

# Tickers
tickers = {
    "NIFTY 50": "^NSEI",
    "Dow Futures": "YM=F",
    "Nasdaq Futures": "NQ=F"
}

# Download 5 years of data
data = {}

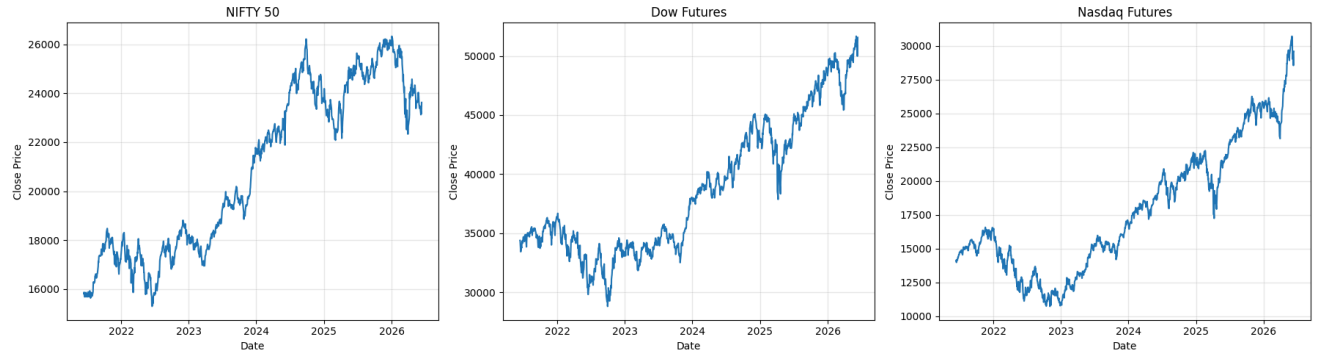
for name, ticker in tickers.items():
    df = yf.download(
        ticker,
        period="5y",
        auto_adjust=True,
        progress=False
    )
    data[name] = df

# Create side-by-side charts
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, (name, df) in zip(axes, data.items()):
    ax.plot(df.index, df["Close"], linewidth=1.5)
    ax.set_title(name)
    ax.set_xlabel("Date")
    ax.set_ylabel("Close Price")
```

```
ax.grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.show()
```



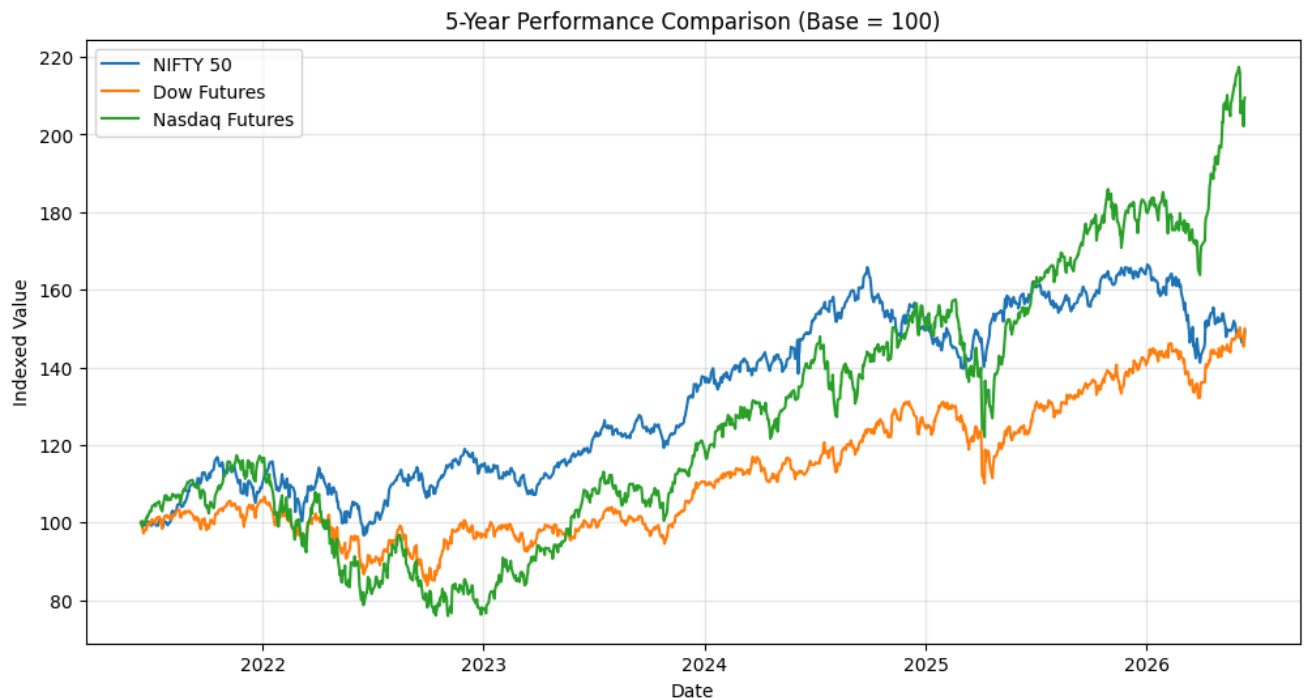
```
# Normalized comparison (base = 100)
```

```
fig, ax = plt.subplots(figsize=(12, 6))
```

```
for name, df in data.items():
    normalized = (df["Close"] / df["Close"].iloc[0]) * 100
    ax.plot(normalized.index, normalized, label=name)
```

```
ax.set_title("5-Year Performance Comparison (Base = 100)")
ax.set_xlabel("Date")
ax.set_ylabel("Indexed Value")
ax.legend()
ax.grid(True, alpha=0.3)
```

```
plt.show()
```



```
!pip install yfinance --quiet
```

```
import yfinance as yf
import matplotlib.pyplot as plt
```

```
# Indices
tickers = {
    "NIFTY 50": "^NSEI",
    "DAX": "^GDAXI",
    "FTSE 100": "^FTSE",
    "China A50": "CN=F"
}

data = {}

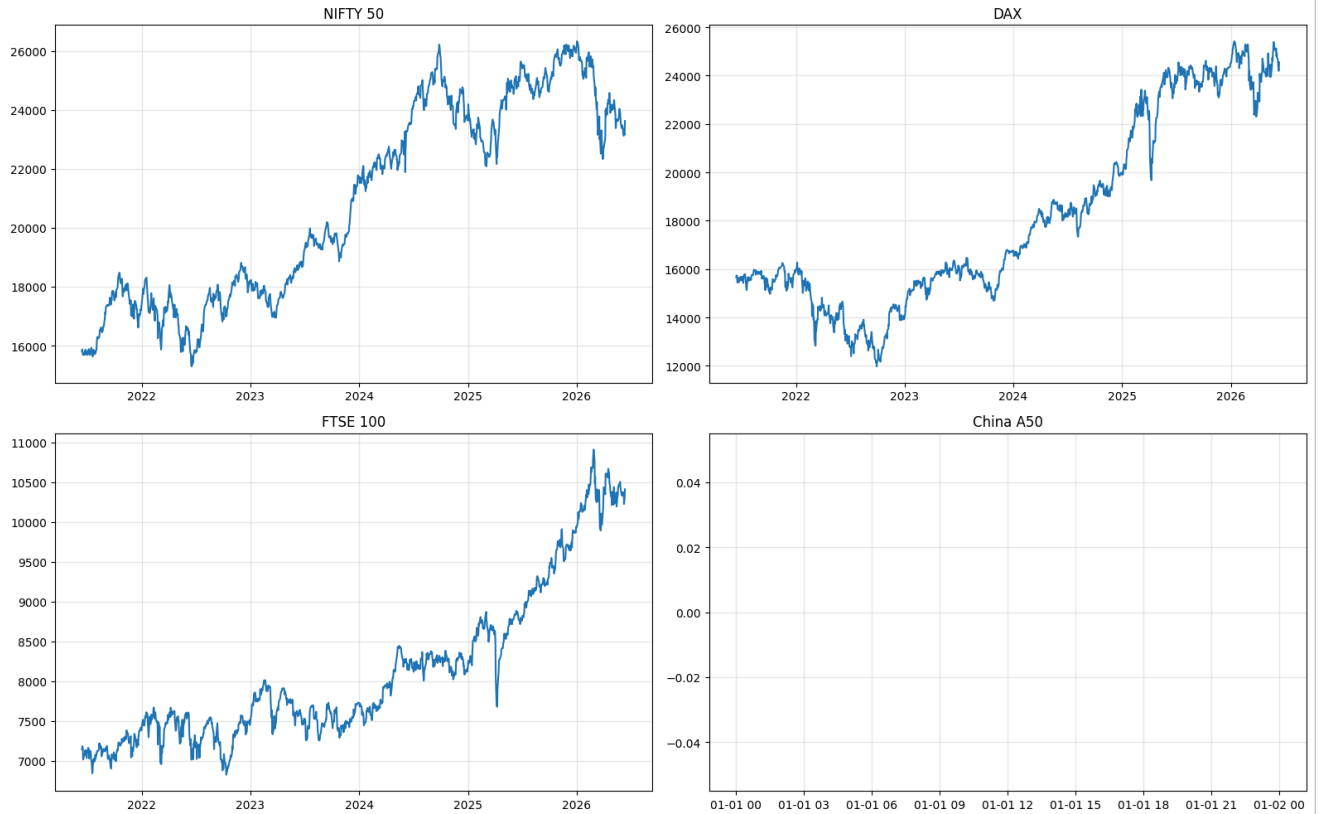
for name, ticker in tickers.items():
    df = yf.download(
        ticker,
        period="5y",
        auto_adjust=True,
        progress=False
    )
    data[name] = df

# 2x2 grid
fig, axes = plt.subplots(2, 2, figsize=(16, 10))
axes = axes.flatten()

for ax, (name, df) in zip(axes, data.items()):
    ax.plot(df.index, df["Close"], linewidth=1.5)
    ax.set_title(name)
    ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```

ERROR:yfinance:HTTP Error 404: {"quoteSummary":{"result":null,"error":{"code":"Not Found","description":"Quote not found for sym
 ERROR:yfinance:
 1 Failed download:
 ERROR:yfinance:['CN=F']: YFPricesMissingError('possibly delisted; no price data found (period=5y) (Yahoo error = "No data found



```
!pip install yfinance --quiet
```

```
import yfinance as yf
import matplotlib.pyplot as plt
```

```
tickers = {
    "NIFTY 50": "^NSEI",
    "Nikkei 225": "^N225",
    "KOSPI": "^KS11",
    "Hang Seng": "^HSI"
}
```

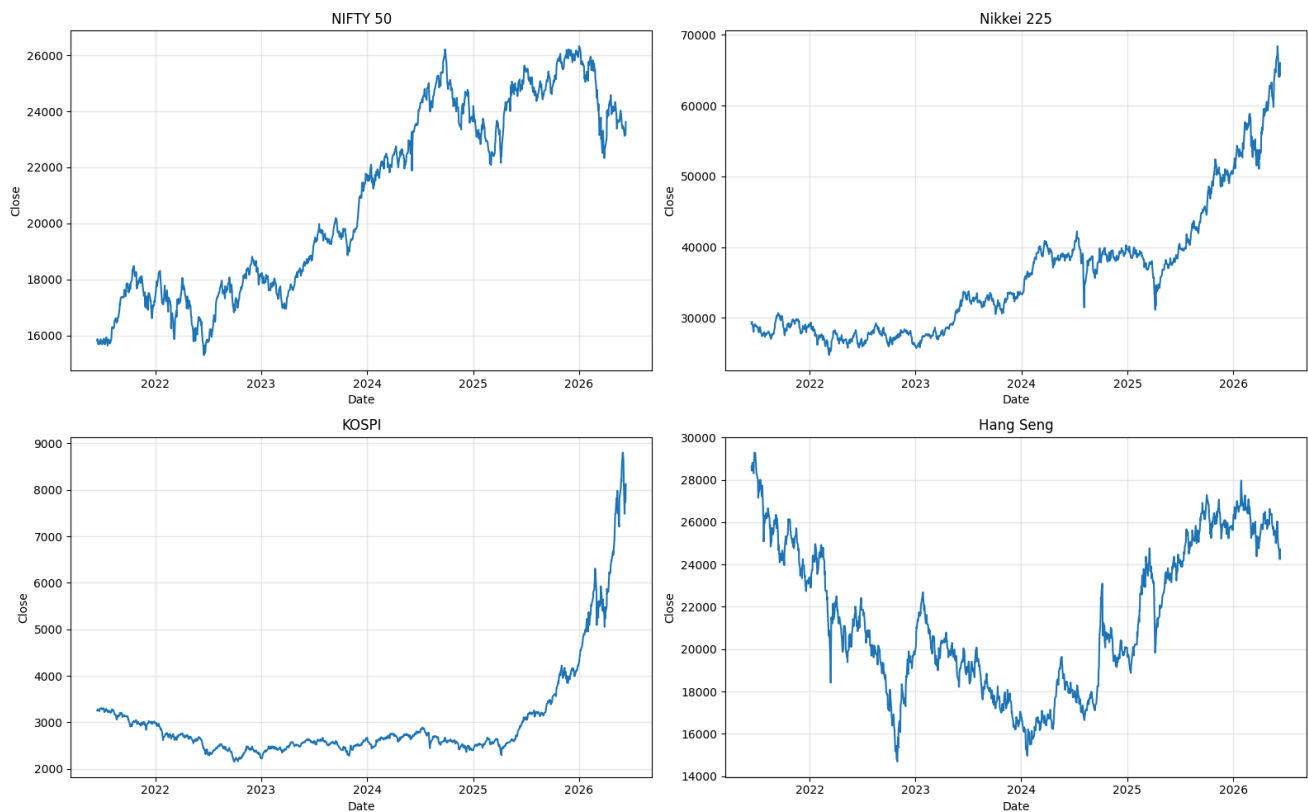
```
data = {}
```

```
for name, ticker in tickers.items():
    df = yf.download(
        ticker,
        period="5y",
        auto_adjust=True,
        progress=False
    )
    data[name] = df
```

```
# Side-by-side charts (2x2)
fig, axes = plt.subplots(2, 2, figsize=(16, 10))
axes = axes.flatten()
```

```
for ax, (name, df) in zip(axes, data.items()):
    ax.plot(df.index, df["Close"], linewidth=1.5)
    ax.set_title(name)
    ax.set_xlabel("Date")
    ax.set_ylabel("Close")
    ax.grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.show()
```



```
!pip install yfinance --quiet
```

```
import yfinance as yf
import matplotlib.pyplot as plt
```

```
tickers = {
    "NIFTY 50": "^NSEI",
    "Gold": "GC=F",
    "Silver": "SI=F"
}
```

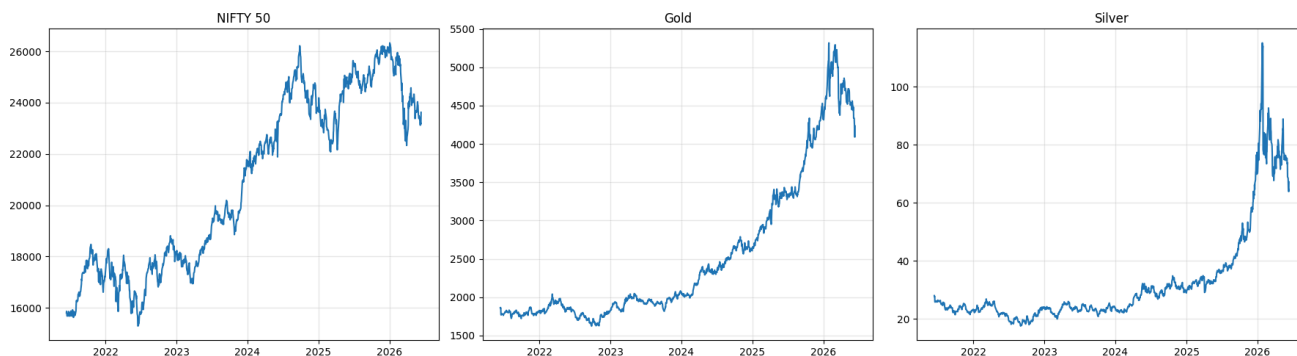
```
data = {}
```

```
for name, ticker in tickers.items():
    df = yf.download(
        ticker,
        period="5y",
        auto_adjust=True,
        progress=False
    )
    data[name] = df
```

```
# Side-by-side charts
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, (name, df) in zip(axes, data.items()):
    ax.plot(df.index, df["Close"])
    ax.set_title(name)
    ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



```
!pip install yfinance --quiet

import yfinance as yf
import matplotlib.pyplot as plt

tickers = {
    "NIFTY 50": "^NSEI",
    "Dollar Index (DXY)": "DX-Y.NYB"
}

data = {}

for name, ticker in tickers.items():
    df = yf.download(
        ticker,
        period="5y",
        auto_adjust=True,
        progress=False
    )
    data[name] = df

# Side-by-side charts
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

for ax, (name, df) in zip(axes, data.items()):
    ax.plot(df.index, df["Close"])
    ax.set_title(name)
    ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



```
!pip install yfinance --quiet

import yfinance as yf
import matplotlib.pyplot as plt

tickers = {
    "NIFTY 50": "^NSEI",
    "WTI Crude": "CL=F",
    "Brent Crude": "BZ=F"
}

data = {}

for name, ticker in tickers.items():
    df = yf.download(
        ticker,
        period="5y",
        auto_adjust=True,
        progress=False
    )
    data[name] = df

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, (name, df) in zip(axes, data.items()):
    ax.plot(df.index, df["Close"])
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    ax.grid(True, alpha=0.3)

plt.tight_layout()
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